Semantic Representation in Customer Support Chatbots

```
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe "c:/U
sers/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/2.py"
 Q1
Actual    : ACTIVATE(Roaming, Customer)
Predicted : ACTIVATE(Roaming, Customer)
Result    : Correct

 Q2
Actual    : DEACTIVATE(CallerTune, Customer)
Predicted : ACTIVATE(CallerTune, Customer)
Result    : Semantic Error

 Q3
Actual    : QUERY(DataBalance, Customer)
Predicted : QUERY(DataBalance, Customer)
Result    : Correct

 Q4
Actual    : ACTIVATE(5GService, Customer)
Predicted : ACTIVATE(5GService, Customer)
Result    : Correct

Total Queries : 4
Correct       : 3
Incorrect     : 1
Accuracy      : 75.0 %
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)>
```

2. First-Order Predicate Calculus for Smart Manufacturing

```
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe "c:/U
sers/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/4.py"
SMART MANUFACTURING
-------------------
Active( M1 ) -> Producing( M1 )
Active( M2 ) -> Producing( M2 )
Maintenance( M3 ) -> NOT Producing( M3 )
Active( M4 ) -> Producing( M4 )

PRODUCT AVAILABILITY
--------------------
Gear -> Available
Shaft -> Available
Gear -> Not Available
Bolt -> Available

GEAR PRODUCTION
----------------
Gear production is affected.
Reason: M3 produces Gear but M3 is under maintenance.
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)>
```

3. Word Sense Disambiguation in E-Commerce

```
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe "c:/U
sers/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/5.py"
------------------------

Query          : Apple accessories
Clicked Result : iPhone Charger
Semantic Cue   : iPhone
Correct Sense  : Technology Brand

Query          : Mouse wireless
Clicked Result : Bluetooth Mouse
Semantic Cue   : Bluetooth
Correct Sense  : Computer Device

Query          : Java tutorial
Clicked Result : Coding Lessons
Semantic Cue   : Coding
Correct Sense  : Programming Language

Query          : Python course
Clicked Result : Software Development Training
Semantic Cue   : Software Development
Correct Sense  : Programming Language

CONCLUSION
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All 4 queries are correctly disambiguated.
```

4. Syntax-Driven Semantic Analysis in Healthcare

```
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe "c:/U
sers/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/7.py"
Doctor -> Agent
Medicine -> Theme
Patient -> Recipient

Sentence: Patient reported severe headache.
Semantic Roles:
Patient -> Experiencer
Headache -> Symptom

Sentence: Nurse monitored patient continuously.
Semantic Roles:
Nurse -> Agent
Patient -> Theme

Sentence: Medicine reduced blood pressure.
Semantic Roles:
Medicine -> Cause
Blood Pressure -> Affected Entity

ROLE ERROR ANALYSIS
-------------------
Given role for Medicine : Instrument
Correct role            : Theme
Error: Incorrect semantic role detected.
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)>
```